

Faculty Profile

Name: Dr. REKHA TRIPATHI
 Designation: Associate Professor
 Qualification: Ph.D, M.Sc
 Mobile No.: 9958735457
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Area of Interest/Specialization: Phytochemistry, Pharmacology, Environmental studies

Experience: 27+years

Educational Qualification

S No.	Degree/Diploma/Certificate or other specialised education obtained	College and/ or University	Year	Percentage	Division
1.	Ph.D.	University of Rajasthan, Jaipur	1996		
2.	M. Sc.	University of Rajasthan, Jaipur	1991	70%	First
3.	B. Sc.	Mohanlal Sukhadia University, Udaipur	1989	71%	First
4.	Higher Secondary	Rajasthan Board, Ajmer	1986	75.75%	First
5.	Secondary	Rajasthan Board, Ajmer	1985	71%	First
6.	Diploma in Gem Identification	Gem Testing Laboratory	1998	Best Student Award for the year 1998	

Employment Record

S No.	From [Month & Year]	To [Month & Year]	Employer Name	Positions Held	No of Years
1.	June 2017	Till date	MSIT	Associate Professor	8+years
2.	Sept. 2009	May 2017	MSIT	Assistant Professor	8 years
3.	Sept. 2007	April 2008	AIET, Jaipur	Lecturer	8 months
4.	1996 1998	1997 2007	Kanoria Mahila Mahavidyalaya, Jaipur	Temporary Lecturer	10 years
5.	1995	1996	Maharani's College, Jaipur	Temporary Lecturer	1 year
6.	1999	2000	GTL, Jaipur	guest faculty for Diploma Courses	1 year

Key Publications

29 papers published in various SCI, SCIE, ESCI, SCOPUS International Journal and Conference (**Citations: 87, H-index: 7, i10-index: 5**)

Paper Publications

1. R. Tripathi, M. Chaudhary, “Fluoride Variability and Proportionate Risk Assessment Using Monte Carlo Simulation in Groundwater of Dausa District, Rajasthan, India” J. of Chemical Health Risks, 2026 (**Scopus indexed**).
2. R. Tripathi, “A Comprehensive Review on Pharmaceutical Studies of Hibiscus micranthus Linn.” IJRPR, 2026 (**Google Scholar**).
3. R. Tripathi, “Effects of Hazardous Waste on the Environment and Strategic Management Approaches”, IJSRST, 2026 (**UGC Care**).
4. R. Tripathi, M.Chaudhary, S.Singh ,S. Dhull, “ Phytochemical and Antimicrobial Screening of Stem Bark of Thevatia peruviana” Oriental J. of Chemistry,2025 (**E-SCI , Scopus indexed**).
5. R. Tripathi, “Application of Waste Heat Recovery Boiler for Energy Conservation in Lead Smelting in Dariba Mines” International J of Chemistry Research, 2025 (**Google Scholar**).
6. M.Chaudhary, S.Singh ,S. Dhull, R.Tripathi, R. Gupta, “Comprehensive Review of Nanomaterial Synthesis Methods, Characterization and Multifaceted Applications in Medical field” Journal of Polymer & Composites, 2025 (**E-SCI**)
7. R. Tripathi, “Antioxidants: Types, Functions and Usage”, IJSRST, 2024 (**UGC Care**).
8. R. Tripathi, “Comparative Antimicrobial Studies of Aerial parts and Roots of Abutilon bidentatum” IJRSET, 2024 (**Google Scholar**)
9. R. Tripathi, “An Overview of Pharmacological Studies of Mimosa hamata (Willd.) – A Medicinal Plant” Heterocyclic Letters,2023, (**E-SCI**)
10. R. Tripathi, “Seasonal Variation in Ground Water Quality of Baswa Tehsil, Dausa District, Rajasthan: A Statistical Approach”, Environment and Ecology, 2023 (**Web of Sci, Zoological record**) .
11. R. Tripathi & A. Sharma, “Phytochemical and Antimicrobial Analysis of Root Extract of *Asparagus adscendens*”, Oriental J. of Chemistry,2022 (**E-SCI , Scopus indexed**).
12. R. Tripathi, “ Antimicrobial Studies of Metahnolic Extracts of Leaves, Stem bark and Roots of Holoptelea Integrifolia Planch”, IJRSET,2022(**DOAJ Indexed, Google Scholar, Crossref**)
13. R. Tripathi, “Assessment of Fluoride Content in Ground Water and its Impact on Wheat and Vegetables of Dausa District, Rajasthan, India.” IJRASET, 2021(**Research Gate,UGC, Google Scholar**).
14. R. Tripathi, “Antimicrobial Studies of Different Extracts of Holoptelia Integrifolia(ROXB.)Leaves”. AJPCR (Asian J Pharm Clin Res)2020(**Google Scholar, Elsevier Products (EMBASE) Crossref**).

15. R.Tripathi, "ANTIMICROBIAL INVESTIGATION OF SOME ARID ZONE PLANTS OF RAJASTHAN", International Journal of Pharmaceutical Sciences and Research, 2017(**E-SCI , Scopus indexed**).
16. R.Tripathi ,” Antimicrobial Investigation of Aerial parts and Roots of Heliotropium ellipticum Ledeb, European Journal of Biomedical and Pharmaceutical Sciences,2017(**Crossref, Google Scholar,GIF 0.377**)
17. R.Tripathi ,” Pharmacological Study of Aerial Parts of two Malvaceous plants. International Journal of Advanced Research,2017 (**Crossref, Google Scholar, Pubmed**)
18. R.Tripathi ,” Phytochemical and Pharmacological Studies of Abutilon bidentatum Hochst.,International Journal of Current Pharmaceutical Research,2017 (**Elsevier Products,Google Scholar,SCI mago**)
19. R. Tripathi, “Study of Heat Recovery from Waste Heat Recovery Boiler in Dariba Lead Smelter” International Journal for Scientific Research & Development(**Research Gate, Google Scholar**)
20. R. Tripathi “Comparative pharmacological study of Aerial parts and roots of ethanolic extract of Hibiscus micranthus Linn” Journal of Medical Pharmaceutical and Allied Sciences(JMPAS),2016 (**Scopus, Crossref, Google Scholar**)
21. R. Tripathi, “Study of Tail Gas Treatment in Dariba Lead Smelter” Int. Res. J. Environment Sci,2017 (**Google Scholar,CAS**)
22. R. Tripathi ,”Pharmacological Investigation of Ethanolic Extract of Roots of Holoptelea integrifolia (Roxb.)Plench”, IJRASET(**DOAJ Indexed, Google Scholar, Crossref**)
23. R. Jain, S.Alam, R. Arora and S.C. Jain, “Phytochemistry and Bioactivity of Kirganelia reticulata”, Journal of Medicinal and Aromatic Plant Sciences,1998 (**Scopus indexed**).
24. R. Jain and R. Arora, “ Phytochemical and Pharmacological Investigation of Heliotropium ellipticum Ledeb.” ,J. Indian Chem. Soc.,1997 (**SCI indexed , Scopus, UGC Care list, Impact factor – 0.284**)
25. R. Jain and R. Arora, “3-O-L Arabinosyl-D-glucosyl morolic acid from Mimosa hamata Willd.” Journal of Medicinal and Aromatic Plant Sciences, 1997 (**Scopus indexed**).
26. R. Jain, R. Arora and S.C. Jain, “Chemical Constituents and Bioactivity Studies of Hibiscus micranthus Linn.”, Indian J. Pharmaceutical Sciences, 1997(**SCI indexed , Scopus, UGC Care list, Impact factor – 0.97**)
27. R. Jain and R. Arora, “Saponin from the roots of Mimosa hamata Willd.”, Indian J. Chem.1997 (**SCI indexed , Scopus, UGC Care list, Impact factor – 0.471**)
28. R. Jain and R. Arora, S. Alam and S.C. Jain, “Pharmacological Evaluation of Roots of Mimosa hamata”, Fitoterapia,1997 (**SCI indexed , Scopus, Impact factor – 3.204**)
29. R. Jain, S.C. Jain and R. Arora, “A new cholestane derivative of Abutilon bidentatum Hochst and it’s Activity”, Die pharmazie, 1996 (**SCI indexed , Scopus, Impact factor – 2.59**)

Patent/Copyright

1. **Dr. Rekha Tripathi**, Dr. Rinky Dwivedi, Dr. Koyel Datta Gupta, Dr. Tripti Sharma, Dr. Archana Balyan, Dr. Meena Rao “Method for authenticating and validating quick response code” published on 9April,2021 with application no. 202121010704 A.
2. Maharaja Surajmal Institute of Technology, Prof. Archana Balyan, Prof. Tripti Sharma, **Dr. Rekha Tripathi**, Dr. Meena Rao, Dr. Bharti Sharma, Dr. Richa Gupta, Dr. Priya Dalal, Ms. Priyanka, Dr. Parul Chaudhary “ARTIFICIAL INTELLIGENCE, BIG DATA, INTERNET OF THINGS, AND BLOCKCHAIN TECHNIQUES AND APPLICATIONS IN SMART ENERGY MANAGEMENT Published on 19/01/2024 with application no. 202311087301.
3. Dr. Malvika Chaudhary, Dr. Sobinder Singh, Dr. Rekha Tripathi, Dr. Anjali, Dr. Sachin Dhull, Ms. Reenu Kumari, Dr. Rashmi Gupta, Dr. Swati Chaudhary, Dr. Gitanjali “IOT ENABLED CHEMICAL MIXING DEVICE FOR AGRICULTURE FARM ANALYSING”. Published on 3/10/2025 with application no. 202511086306 A

Research Papers presented in Conferences/ Seminars

S No.	Title of Paper	Name of Conferences/ Seminars	National/ International	Name of Host Institution	Year
1.	Chemical Toxicology in Environment and Public Health: Risk Assessment, Regulation & Emerging Issues	NCRTCCES 2026	National	MSIT, 27-28 February,	2026
2.	Digital Distraction and Its Effect on Attention Span among Engineering Students	International Conference of Advance Research and Innovation (ICARI-2026)	International Conference of Advance Research and Innovation (ICARI-2026)	Delhi State Center, Institution of Engineers, Delhi on 2 February,	2026
3	A Review on Bioactive constituents of Medicinal Plants	ICMPHSH International Conference on Multidisciplinary Perspective in Science & Humanities Recent Trends & Innovation - 2025	International	16" to 17" May, 2025 at Arya College of Engineering &I.T. (Arya 1st Old Campus) Kukas, Jaipur	2025
4	Mechanical Aspects of AI algorithms and Mathematical and experimental Investigation of Hybrid Abrasive Flow Machining Process	International Conference of Artificial Intelligence and Application ICAIA 2025	International	MSIT, 26th, 27th March 2025	2025
5	“A Comprehensive Review on Climate Change Mitigation Strategies and Their Impact on Achieving Sustainability Goals”	International Conference of Advance Research and Innovation (ICARI-2025)	International Conference of Advance Research and Innovation (ICARI-2025)	Delhi State Center, Institution of Engineers, Delhi on 2 February,	2025

6	From Tradition to Innovation: Artificial Intelligence and machine Learning in Molecular Docking	Proceedings of NCRTCES-2024	National Conference on Recent Trends in Climate Change, energy, and Sustainability	MSIT, 11-12 November,	2024
7	Ecological conservation, biodiversity, environmental laws, pollution and environmental impact assessment in India: A brief overview	Proceedings of NCRTCES-2024	National Conference on Recent Trends in Climate Change, energy, and Sustainability	MSIT, 11-12 November,	2024
8	A Review on Climate Change and Sustainability	Proceedings of NCRTCES-2024	National Conference on Recent Trends in Climate Change, energy, and Sustainability	MSIT, 11-12 November,	2024
9	A new Saponin from the roots of Mimosa hamata Willd	Indian Chemical Society	National	Univ. Of Rajasthan	1995
10	Phytochemical Studies of Hibiscus micranthus Linn.	Indian Science Congress	National	Univ. Of Rajasthan	1993

Book Published

S No.	Title of Book/Journal	Publisher	National/ International	Volume and Issue no. (in case of journal)	Year of Publication
1	Environmental Chemistry	JAYA Publishing House, Delhi	International	ISBN:978-93-87590-80-9	2019
2	Energy and Environment	Photon eBooks	International	015-A94510112043	2017
3	Air Pollution by Lead Smelter & it's Control	Lambert Academic Publishing,	International	978-3-659-97819-7	2016

MOOC Courses (NPTEL Swayam)

S.No.	Title of Course	Organized by	Duration	Total no. of weeks
1.	Introduction to polymer science	NPTEL	July-Sept. 2023	12 Weeks
2.	Emotional Intelligence	NPTEL	Jan-March 2026	8 Weeks

Trainings and FDPs

S No.	Name of Training	Name of Training Institution, City	Type of Training and FDPs	Period	
				From	To
1.	Mindfulness in Motion: Strengthening Academic Performance Through Meditation	Organized by MSIT	FDP (Offline)	25-05-2026	29-05-2026
2.	Modern Educators AI Tool Box for Productivity Enhancement	through ICT by NITTTR Chandigarh	FDP	18-05-2026	22-05-2026
3.	Bridging Intelligence and Connectivity: Emerging perspectives in the era of AI	AICTE Training And Learning (ATAL) Academy organized by MSI	FDP	15-12-2025	20-12-2025
4.	Developing Healthy Work Culture at Workplace	through ICT by NITTTR Chandigarh	FDP	1-12-2025	5-12-2025
5	NEP 2020 Orientation & Sensitization Programme	Malaviya Mission Teacher Training Programme (MMTTP) of University Grants Commission (UGC)	Training program	4-11-2025	13-11-2025
6.	Effective Teaching - Learning Using Social Media	through ICT by NITTTR Chandigarh	FDP	26-05-2025	30-05-2025
7.	Life Skills Development	through ICT by NITTTR Chandigarh	FDP	27-01-2025	31-01-2025
8.	Developing Healthy Work Culture at Work Place	through ICT by NITTTR Chandigarh	FDP	02-12-2024	06-12-2024
9.	Curriculum Design for Outcome Based Education	through ICT by NITTTR Chandigarh	FDP	03-06-2024	07-06-2024
10.	Energy Storage Materials and Technology	through ICT by NITTTR Chandigarh	FDP	06-05-2024	10-05-2024
11.	Digital Tools for Teaching and Learning	through ICT by NITTTR Chandigarh	FDP	11-03-2024	15-03-2024
12	Education and Educational Management Department	through ICT by NITTTR Chandigarh	FDP	15-01-2024	19-01-2024

13.	Blended Learning and Flipped Classroom	through ICT by NITTTR Chandigarh	FDP	16-10-2023	20-10-2023
14.	Outcome based Curriculum and Accreditation Criteria	through ICT by NITTTR Chandigarh	FDP	03-07-2023	07-07-2023
15.	Soft Skills & Classroom Teaching	through ICT by NITTTR Chandigarh	FDP	20-03-2023	24-03-2023
16.	Team Building , Motivation & Creativity	through ICT by NITTTR Chandigarh	FDP	12-12-2022	16-12-2022
17.	Entrepreneurship, Innovation & Digital Skills in New Era of AI	MIT Muradabad	FDP	25-07-2022	29-07-2022
18.	Skill Development for Entrepreneurs	AICTE,ISTE, organized by MSIT	FDP	27-12-2021	01-01-2022
19.	Use of ICT in Engineering Education	AICTE	FDP	7-04-2021	13-04-2021
20.	Faculty Development: Planning and Management for Institutional Excellence	through ICT by NITTTR	FDP	18-11-19	22-11-19
21.	Classroom Communication	through ICT by NITTTR	FDP	28-05-2018	1-06- 2018
22.	Comprehensive Development of Technical Teachers	HMR Institute of Technology & Management, New Delhi.	FDP	8-05-2017	13-05-2017
23.	Orientation Programme	Jaipur	UGC Sponsored Training	08-06-2005	05-07-2005

Conferences/ Workshops/ Seminars/Webinars Attended

S No.	Name of Conferences/ Workshops/ Seminars	Name of Institution, City	Type of Conferences/ Workshops/ Seminars	Period	
				From	To
1.	National Intellectual Property Awareness Mission	In MSIT by intellectual property, INDIA	Training Program	12-03-2022	
2.	Infrared Spectroscopy: Fundamentals & Applications	Deshbandhu College	Seminar	8-04-2021	
3.	ADMET	MSIT	Conference	5-03-.2021	6-03-2021
4.	Ancient Indian Education System	NITTTR,Kolkata	Conference	11-11-2020	12-11-2020

5.	Artificial intelligence and applications	MSIT	International Conference	6 th and 7 th Feb.. 2020	
6.	Managing Self and Learning during Covid-19 Pandemic	MSI	Webinar	08-10-2020	
7.	National Education Policy 2020	MSIT	Webinar	22-09-2020	
8.	National Pension Scheme :Pre and Post Covid scenario in India	MSI	Webinar	08-08-2020	
9.	Covid 19: It's impact on Engineering Disciplines	MSI	Webinar	07-08-2020	
10.	"Life Enhancement Skills post COVID-19 Pandemic"	MSI	Webinar	29-07-2020	
11.	The Future of Online Teaching in India - Challenges & Solutions	Business Stand Learning	Webinar	24-07-2020	
12.	Artificial Intelligence with Machine Learning	MSI	Webinar	22-07-2020	
13.	ICAIA	MSIT	International conference	6-02-2020	7-02-2020
14.	Single Atomic Layer based Semiconductor Devices and Research Opportunities in South Korea	MSIT	Seminar	3-10-2019	
15.	Paradigm Shift from Developing to Developed India through- Digitalization	MSI	National conference	28-07-2017	29-07-2017
16.	Signal and Noise during Electronic Measurements	MSIT	Seminar	27-02-2016	
17.	Climate change	MSIT	Seminar	20-04-2013	
18.	Plasma and its own world	MSIT	Seminar	08-03-2013	
19.	Positive linear operators	MSIT	Seminar	13-02-2013	
20.	Science For Shaping Future Of India	Indian Science Congress Delhi Chapter, Delhi	Seminar	26-11-2012	27-11-2012
21.	Nanotechnology	MSIT	Seminar	16-10-2012	

Membership of Professional Bodies

S No.	Name of the Professional Body	Member/Fellow	Remarks
<u>1</u>	Indian Science Congress	Member , ID – L 16817	<u>Life time member</u>

Other Contributions

- **Head of the Department** from **Sept.2023 to May 2026.**
- **Convener Annual day 2024**
- **Convener Annual day 2023**
- **Convener** of Student Insurance & Institute building / Lift Committee from Sept. 2021
- **Convener** GGSIPU Academic Audit in 2019-2020 & 2020-2021
- **Head of the Department** from 2013- Nov 2020
- **Member** internal complaint committee since 2017
- **Member** Discipline / anti ragging committee from 2017-2019
- **Convener** Library committee from 2017-2019
- **Co-convener** Discipline / anti ragging committee 2016-2017
- **Co-convener** exam cell 2016-2017
- **Member** Sexual harassment cell 2016-2017
- Core committee member International Conference organized in MSIT **ICAIA 2020.**
- Core committee member Conference organized in MSIT **AdMet-2021.**

Dr. Rekha Tripathi

Associate Professor

Applied Science Department

Maharaja Surajmal Institute of Technology